

WHAT DOES $\epsilon = 1$ ACTUALLY MEAN IN PRACTICE BUILDING A REGISTRY TO FIND OUT

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1: Augusta Griffin, 2: NIST, 3: Working Paper

As differential privacy moves from academic research to real-world deployment in census data, advertising technology, and healthcare applications, practitioners face a persistent challenge: there are few systematic resources to learn from others' implementation decisions. Privacy engineers designing new systems often rely on fragmented blog posts, informal spreadsheets, and anecdotal guidance when making critical choices about privacy parameters, deployment models, and trust assumptions.

This talk presents a new NIST resource currently in development: a community-driven Differential Privacy Deployment Registry¹ designed to systematically collect, organize, and disseminate information about real-world deployments of differential privacy. We describe how public documentation of privacy implementations can serve three interconnected audiences: practitioners seeking to benchmark their design choices against existing deployments, policymakers evaluating whether new systems meet emerging standards, and researchers studying ecosystem-level patterns in privacy practice.

We will discuss the registry's technical schema—covering privacy variants, privacy units, epsilon values, deployment models, and implementation details—and explain how a tiered transparency framework allows entries to be compared even when organizations disclose different levels of detail. We also address the governance and editorial processes needed to ensure submissions are reliable while remaining accessible to community contribution.

Drawing on examples from deployments by Apple, Google, Microsoft, the U.S. Census Bureau, and others already documented in the registry prototype, we demonstrate how centralized documentation of privacy choices can accelerate industry consensus, inform regulatory guidance, and create virtuous cycles of transparency. We conclude with reflections on how registries and benchmarking resources can complement formal privacy standards—such as NIST SP 800-226's guidelines for evaluating differential privacy guarantees—to support the broader goal of responsible privacy-preserving data use.

¹<https://csrc.nist.gov/pubs/ir/8588/ipd>